Library Vans: Test Two Cost Rules

Tracking #: 2026-11-14-NAT-MATH-VAN-RULES-47

Invented community library-van prices, not a real transport quote

Plan	Starting fee	Cost per mile
Pine	$8	$5
Cedar	$20	$3
Distance examples	2, 6, 8 miles	Same miles for both

1. Build a table for both plans at 0, 2, 4 and 6 miles. Show the fixed starting fee when distance is zero, not a zero total.

2. Choose 4 miles and substitute it into both rules. Calculate every step and say exactly which plan costs less and by how much.

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3. Show complete substitutions proving the tie at 6 miles. Explain in TWO sentences why that does not mean both plans cost the same for all distances.

4. Check 8 miles. Give both full equations and identify which plan wins and by how many dollars. Use this as a counterexample to ’Pine is always cheaper.’

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5. A traveler expects to drive 5 miles and has a $35 budget. Compute both plans, show whether each fits, and state exactly how much money remains or is missing.

6. Write a short buyer recommendation for a 2-mile trip and an 8-mile trip. Include both numerical costs and an accurate limit on your recommendation.